Reg. No										

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
MT203 - CALCULUS

Time : 3 hrs **Max Marks :75**

SECTION - A (5X2=10)

Answer **ALL** Questions

1. If $xy = c^2$ find $\frac{dy}{dx}$.
2. Write the Cartesian formula for radius of curvature.
3. Find the p-r equation of $r = a\sin\theta$.
4. Evaluate $\int \sec^3 x dx$.
5. Evaluate $\int_0^2 \int_{x^2}^{2x} (2x+3y) dy dx$.

SECTION - B (3X5=15)

Answer any **THREE** Questions

6. Find the minimum value of $x^2 + y^2 + z^2$ where $ax + by + cz = p$.
7. Find the curvature at any point of the curve $x = ae^{\theta}(\sin\theta - \cos\theta), y = ae^{\theta}(\sin\theta + \cos\theta)$.
8. Find the envelope of the family of straight line $y = mx + \frac{a}{m}$, the parameter being t.

9. Show that $\Gamma(n+1) = n\Gamma(n) = n!$.

10. Evaluate $\int_0^{\pi/2} \int_0^{a\cos\theta} r\sqrt{a^2 - r^2}\, dr d\theta$.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) If $y = \left(x + \sqrt{1+x^2}\right)^m$ prove that
$\left(1+x^2\right) y_{n+2} + (2n+1)\, x y_{n+1} + \left(n^2 - m^2\right) y_n = 0.$

(or)

b) If $x = v^2 + w^2, y = w^2 + u^2, z = u^2 + v^2$ show that
$\frac{\partial(x,y,z)}{\partial(u,v,w)} \cdot \frac{\partial(u,v,w)}{\partial(x,y,z)} = 1.$

12. a) Find the radius of curvature of $y^2 = \frac{a^3 - x^3}{x}$ at $(a,0)$.

(or)

b) Derive the formula for radius of curvature in polar coordinates.

13. a) Find the evolute of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

(or)

b) Find the rectilinear asymptotes of
$2x^4 - 5x^2y^2 + 3y^4 + 4x^3 - 6y^3 + x^2 + y^2 - 2xy + 1 = 0.$

14. a) Express $\int_0^1 x^m \left(1-x^n\right)^p dx$, in terms of gamma functions and evaluate the integral $\int_0^1 x^5 \left(1-x^3\right)^{10} dx$.

(or)

b) If $In = \int_0^{\frac{\pi}{2}} x \cos^n x dx$ where $n > 1$ show that $I_n = \frac{-1}{n^2} + \frac{n-1}{n} I_{n-2}$. Find I_4.

15. a) Find the area common to $y^2 = 4ax, x^2 = 4ay$

(or)

b) Evaluate $\int_0^{\log 2} \int_0^x \int_0^{x+\log y} e^{x+y+z} dz dy dx$.

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